A FINITE PURELY INSEPARABLE EXTENSION IS AN ITERATED TOWER OF P-TH ROOT ADJUNCTIONS OF DEGREE P

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The tower produced here is the input for the purely inseparable case of lemma-separable-degree (delivered L48344).

Editorial note. The label lemma-finite-purely-inseparable names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-nr-roots-unchanged (the number of p-th roots of an element is unchanged under field extension - used to see t = alpha^p has no p-th root in F), lemma-take-pth-root (adjoining a p-th root of an element with no p-th root gives a degree p purely inseparable extension); definition of purely inseparable (every element has a p-power power in the base)

Source reference. Original reference 09H4: sources/stacks-fields/pinned-original-source.tex, source lines 1442–1442. Prerequisite pointer invoked at 1911; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09HF: sources/stacks-fields/pinned-original-source.tex, source lines 1834–1834. Prerequisite pointer invoked at 1913; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let E/F be a finite purely inseparable field extension of characteristic $p > 0$. Then there exists a sequence of elements $\alpha_1, \ldots, \alpha_n \in E$ such that we obtain a tower of fields*

$$E = F(\alpha_1, \ldots, \alpha_n) \supset F(\alpha_1, \ldots, \alpha_{n-1}) \supset \ldots \supset F(\alpha_1) \supset F$$

such that each intermediate extension is of degree p and comes from adjoining a pth root. Namely, $\alpha_i^p \in F(\alpha_1, \ldots, \alpha_{i-1})$ is an element which does not have a pth root in $F(\alpha_1, \ldots, \alpha_{i-1})$ for $i = 1, \ldots, n$.

Proof. By induction on the degree of E/F. If the degree of the extension is 1 then the result is clear (with $n = 0$). If not, then choose $\alpha \in E$, $\alpha \notin F$. Say $\alpha^{p^r} \in F$ for some $r > 0$. Pick r minimal and replace α by $\alpha^{p^{r-1}}$. Then $\alpha \notin F$, but $\alpha^p \in F$. Then $t = \alpha^p$ is not a pth power in F (because that would imply $\alpha \in F$, see Lemma

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09H4 or its proof). Thus $F \subset F(\alpha)$ is a subextension of degree p (Lemma 09HF). By induction we find $\alpha_1, \ldots, \alpha_n \in E$ generating $E/F(\alpha)$ satisfying the conclusions of the lemma. The sequence $\alpha, \alpha_1, \ldots, \alpha_n$ does the job for the extension E/F. $\square$